1. $A_q = \begin{pmatrix} -5 & 2 & 1 \\ -6 & 2 & 2 \\ 6 & -4 & -4 \end{pmatrix}$ $\quad \det(A-\lambda I)=0$

$$\begin{vmatrix} -5-\lambda & 2 & 1 \\ -6 & 2-\lambda & 2 \\ 6 & -4 & -4-\lambda \end{vmatrix} = (5+\lambda)(4+\lambda)(2-\lambda)+24+24+6(\lambda-2)-12(4+\lambda)-$$

$$-8(5+\lambda) = 2\lambda^2+18\lambda+40-\lambda^3-9\lambda^2-20\lambda+48+$$

$$+6\lambda-12-48-12\lambda-40-8\lambda = -\lambda^3-7\lambda^2-16\lambda-12 = \lambda^3+7\lambda^2+16\lambda+12=0$$

$\lambda^2+9\lambda+20$

$\lambda_1=-2^{(2)}$, $\lambda_2=-3$

$\lambda=-2$: $\begin{pmatrix} -3 & 2 & 1 \\ -6 & 4 & 2 \\ 6 & -4 & -2 \end{pmatrix} \Rightarrow$ $\begin{cases} g^3 = 3g^1-2g^2 \\ \begin{pmatrix} g^1 \\ g^2 \\ g^3 \end{pmatrix} = X_1 \underbrace{\begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}}_{V_1} + X_2 \underbrace{\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}}_{V_2} \end{cases}$

$\lambda=-3$: $\begin{pmatrix} -2 & 2 & 1 \\ -6 & 5 & 2 \\ 6 & -4 & -1 \end{pmatrix} \sim \begin{pmatrix} -2 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 2 \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow$

$\Rightarrow \begin{cases} g^3 = -2g^1 \\ g^2 = 2g^1 \end{cases} \Rightarrow \begin{pmatrix} g^1 \\ g^2 \\ g^3 \end{pmatrix} = X_1 \underbrace{\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}}_{V_3}$

2. $A_q = \begin{pmatrix} -38 & -24 & -6 & -6 \\ 60 & 40 & 12 & 12 \\ -10 & -8 & -10 & -2 \\ -70 & -56 & -14 & -22 \end{pmatrix} \Rightarrow \sigma_q = \{-8^{(3)}, -6\}$

$\lambda=-8$: $\begin{pmatrix} -30 & -24 & -6 & -6 \\ 60 & 48 & 12 & 12 \\ -10 & -8 & -2 & -2 \\ -70 & -56 & -14 & -14 \end{pmatrix} \sim \begin{pmatrix} 5 & 4 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow g^4 = -5g^1-4g^2-g^3$

$$\begin{pmatrix} g^1 \\ g^2 \\ g^3 \\ g^4 \end{pmatrix} = X_1 \underbrace{\begin{pmatrix} 1 \\ 0 \\ 0 \\ -5 \end{pmatrix}}_{V_1} + X_2 \underbrace{\begin{pmatrix} 0 \\ 1 \\ 0 \\ -4 \end{pmatrix}}_{V_2} + X_3 \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}}_{V_3}$$

$\lambda=-6$: $\begin{pmatrix} -32 & -24 & -6 & -6 \\ 60 & 46 & 12 & 12 \\ -10 & -8 & -4 & -2 \\ -70 & -56 & -14 & -16 \end{pmatrix} \sim \begin{pmatrix} 16 & 12 & 3 & 3 \\ 30 & 23 & 6 & 6 \\ 5 & 4 & 2 & 1 \\ 35 & 28 & 7 & 8 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -3 & 0 \\ 0 & -1 & 1 & -1 \\ 0 & 4 & 17 & 1 \\ 0 & 0 & -7 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & -3 & 0 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & 21 & -3 \\ 0 & 0 & -7 & 1 \end{pmatrix} \Rightarrow$

$\Rightarrow \begin{cases} g^4 = 7g^3 \\ g^2 = g^3-g^4 = -6g^3 \\ g^1 = 3g^3 \end{cases} \Rightarrow \begin{pmatrix} g^1 \\ g^2 \\ g^3 \\ g^4 \end{pmatrix} = X_3 \underbrace{\begin{pmatrix} 3 \\ -6 \\ 1 \\ 7 \end{pmatrix}}_{V_4}$